

Ìmòlára: Diacritic-Robust Sentiment Classification for Yorùbá Tweets with Classical, Recurrent and Africa-Centric Transformer Models

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NLP Summative Project — Option 3: Text Classification for an African Language

Abstract

We study three-class sentiment classification (negative, neutral, positive) of Yorùbá tweets, a language spoken by tens of millions of people but poorly served by NLP. Using the AfriSenti-SemEval 2023 Yorùbá data (8,522 / 2,090 / 4,515 tweets), we compare a TF-IDF baseline, BiLSTM/BiGRU models with attention over static embeddings, and six fine-tuned Transformers (mBERT, XLM-R, AfriBERTa, AfroXLMR base/large, and LoRA). Two data problems shape the study: the released test split is pre-normalised while train and dev are raw, and 13% of dev tweets duplicate training tweets; we reconstruct the normalisation and select models on de-duplicated (“clean”) dev data. Africa-centric pre-training helps (+3.7 macro-F1 from XLM-R to AfroXLMR-base) and the 126M-parameter AfriBERTa is the strongest Transformer (clean-test macro-F1 0.732), yet no model significantly beats the tuned TF-IDF baseline (0.737); a learning curve shows the pre-training advantage shrinks from 4.8 to 0.9 points as labelled data grows. All models trained on diacritised text lose 6–13 points when users type without tone marks; training on all three written forms (*mixed3*) removes almost all of this loss at no cost, giving the best test models (TF-IDF 0.745, AfriBERTa 0.741, weighted-F1 77.8 / 77.5 vs. 74.1 for the strongest single model reported with the dataset). Error analysis attributes much of the remaining error to negation, idioms, keyword priors and an estimated 10% label noise. The diacritic-robust models are deployed in a public web application.

Project links. **GitHub:** <https://github.com/Hassan-Adelani-Luqman/imolara-yoruba-sentiment>

Live system: <https://imolara-yoruba-sentiment-oafn6nuzsde8db6a9fdahs.streamlit.app/>

Model: <https://huggingface.co/Hassana-delani1/imolara-afriberta-mixed3>

Demo video: <https://youtu.be/bsjjJIeKi5s>

1 Introduction

Sentiment analysis of social-media text supports public-opinion monitoring, customer-feedback triage and social research. For most African languages such tools hardly exist: Yorùbá, spoken by roughly 50 million people in Nigeria, Benin and the diaspora (Eberhard et al., 2025), has a small fraction of the labelled resources of high-resource languages (Joshi et al., 2020). The NaijaSenti and AfriSenti projects (Muhammad et al., 2022, 2023a) created the first sizeable annotated Yorùbá Twitter corpus and a shared task (Muhammad et al., 2023b), so the task itself is established; what is missing is (i) a systematic, statistically careful comparison of classical, recurrent and Transformer approaches on this data, (ii) an account of how models behave under the way Yorùbá is actually typed, and (iii) a usable tool.

Written Yorùbá marks tone with grave and acute accents and vowel quality with an under-dot (*e*, *o*, *ɔ*); in electronic text these diacritics are frequently omitted (Orife, 2018), which makes words ambiguous (*èdè* ‘language’ vs. *edé* ‘crayfish’, both written *ede*) and changes model behaviour (Adelani et al., 2021). Most AfriSenti Yorùbá tweets are diacritised (77%, §3), but a deployed classifier will meet input from phone keyboards without them.

Objective and research questions. We build and deploy a three-class Yorùbá tweet sentiment classifier and ask: **RQ1** Do Africa-centric pre-trained models outperform general multilingual models and classical baselines? **RQ2** How robust are models to tweets written without tone marks or without any diacritics, and can training fix it? **RQ3** Does training data from related Nigerian languages (Hausa, Igbo, Pidgin) help?

Contributions. (1) We identify and correct a train/test format mismatch in the released data and quantify train/dev/test duplication; (2) we compare more than 30 model configurations, with three seeds for every neural model, bootstrap confidence intervals and paired significance tests; (3) we show that diacritic augmentation makes models robust to undiacritised input at no cost; (4) we analyse errors with reproducible evidence (keyword slices, duplicate-label conflicts, confident learning); (5) we deploy the models in a public web application with a validated int8 ONNX model.

2 Related Work

African-language sentiment. NaijaSenti (Muhammad et al., 2022) annotated tweets in Hausa, Igbo, Nigerian Pidgin and Yorùbá; AfriSenti (Muhammad et al., 2023a) extended this to 14 languages and reported single-model baselines, the best for Yorùbá being AfroXLMR-large (74.1 weighted-F1) and AfriBERTa-large (72.9). In the SemEval-2023 shared task (Muhammad et al., 2023b), the best Yorùbá systems reached about 80 weighted-F1, typically with additional language- and task-adaptive pre-training and ensembling (Wang et al., 2023). These works establish strong numbers but do not study robustness to diacritic variation, the train/test format difference, or duplicate leakage, and they do not provide an end-user tool.

Pre-trained models for African languages. General multilingual encoders such as mBERT (Devlin et al., 2019) and XLM-R (Conneau et al., 2020) cover few African languages; XLM-R’s CC-100 pre-training data contains no Yorùbá. AfriBERTa (Ogueji et al., 2021) is trained from scratch on under 1 GB of text in 11 African languages; AfroXLMR (Alabi et al., 2022) adapts XLM-R to 17 African languages by multilingual adaptive fine-tuning; SERENGETI (Adebara et al., 2023) covers 517 African languages. Comparing XLM-R and AfroXLMR, which share architecture and tokenizer, isolates the effect of African pre-training data (RQ1).

Diacritics in Yorùbá NLP. Orife (2018) frames missing diacritics as a restoration task; Adelani et al. (2021) show diacritics affect Yorùbá–English translation quality. We instead make classifiers *invariant* to diacritic variation by data augmentation, a simpler alternative to a restoration pipeline.

Split	Tweets	Negative	Neutral	Positive
Train	8,522	1,872	3,108	3,542
Dev	2,090	443	763	884
Test	4,515	981	1,616	1,918

Table 1: AfriSenti Yorùbá splits. *Clean* subsets without duplicates of training tweets: 1,817 dev and 4,129 test tweets. (The dataset paper’s counts include the TSV header line.)

Models and evaluation. Our recurrent models follow LSTMs (Hochreiter and Schmidhuber, 1997), GRUs (Cho et al., 2014) and additive attention (Bahdanau et al., 2015) over Word2Vec (Mikolov et al., 2013) or fastText vectors (Bojanowski et al., 2017; Grave et al., 2018). For parameter-efficient fine-tuning we use LoRA (Hu et al., 2022). Fine-tuning large encoders is known to be unstable across seeds (Mosbach et al., 2021), which motivates our multi-seed protocol; significance is assessed with the paired bootstrap (Koehn, 2004; Efron and Tibshirani, 1993), and label noise with confident learning (Northcutt et al., 2021).

3 Dataset and Data Preparation

Source and size. We use the Yorùbá portion of AfriSenti-SemEval 2023 (Muhammad et al., 2023a; AfriSenti-SemEval organisers, 2022) (CC BY 4.0), whose tweets were annotated by three native speakers each with majority voting; free-marginal agreement for Yorùbá is $\kappa = 0.65$. We download the raw TSVs pinned to a fixed repository commit (the Hugging Face copy relies on a loading script that current libraries no longer execute). Table 1 gives the official splits. Negative is the minority class (22%), so macro-F1 is our primary metric.

Exploratory analysis. Tweets are short (median 18 tokens after normalisation; 95th percentile 45). 77% contain tone marks and 73% under-dots, and diacritic presence correlates with the label (85% of neutral vs. 71% of negative tweets carry tone marks), a potential shortcut cue. A heuristic English detector flags about 14% of tweets as code-switched. Test is lexically further from train than dev: 41% of test word types are unseen in training (dev: 29%); ignoring diacritics shrinks the vocabulary by about 30% and lowers the token OOV rate from 7.8% to 5.2%.

A train/test format mismatch. The released test split is pre-normalised (0% uppercase, punctua-

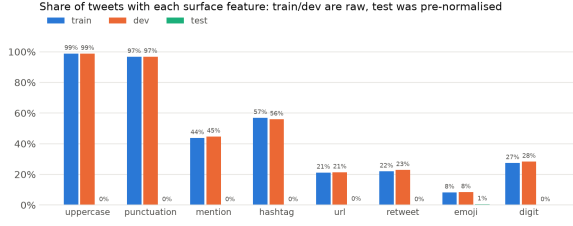


Figure 1: Share of tweets with each surface feature: train/dev are raw, the released test split was pre-normalised.

tion, hashtags or URLs, 0.5% emojis) while train and dev are raw tweets (e.g. 99% uppercase, 57% hashtags; Figure 1). We reconstruct the organisers’ normalisation—lower-casing and removing mentions, URLs, RT, hashtags (with their word), apostrophes, digits, punctuation and emojis, keeping letters and combining marks—and validate it on train/test pairs that are the same tweet: it reproduces 76 of 79 test strings exactly (the three misses differ in the underlying tweet). We apply it, after Unicode NFC normalisation, to every split, in training and in the deployed application.

Duplicates and leakage. Comparing tweets after normalisation and diacritic removal, 13.1% of dev (273) and 8.5% of test tweets (386) duplicate a training tweet—retweets, reposted proverbs, or the same tweet with different hashtags. Duplicates inflate dev scores by about 3 points and bias model selection towards memorisation. We keep the official splits for comparability but report every result on the full split and on a *clean* subset without duplicates, and select models on **clean dev**.

Diacritic forms and auxiliary data. For RQ2 every tweet can be rendered in three forms: *original*, *no tones* (combining grave, acute and macron removed; under-dots kept) and *no diacritics* (all combining marks removed). For RQ3 we use the AfriSenti Hausa (14,172), Igbo (10,192) and Pidgin (5,121) training sets; Pidgin is only 1.4% neutral. No synthetic or LLM-generated data is used.

4 Methodology

Baselines (E0–E1). A majority-class predictor, and TF-IDF features (sublinear tf, $\min_df = 2$) of word 1–2-grams, character 2–5-grams (char_wb) or their union, with logistic regression or a linear SVM (Pedregosa et al., 2011). C and class weighting are grid-searched on clean dev. Word tokens are split on whitespace: scikit-learn’s default to-

ken pattern drops one-letter words such as *o* and breaks words at combining tone marks ($\grave{o}r\acute{e} \rightarrow r\acute{e}$).

Recurrent models (E2). Tokens are embedded (300-d), passed through a bidirectional LSTM or GRU (hidden size $h = 128$ per direction) giving states $H = [\mathbf{h}_1, \dots, \mathbf{h}_T]$, pooled by additive attention (Bahdanau et al., 2015),

$$\alpha_t = \text{softmax}_t \left(\mathbf{v}^\top \tanh(W\mathbf{h}_t) \right),$$

$$\mathbf{c} = \sum_t \alpha_t \mathbf{h}_t,$$

and classified by a linear layer over $\mathbf{c}$ (ablation: concatenated final states). Embeddings are random, pre-trained fastText Yorùbá vectors (Grave et al., 2018), or our own skip-gram Word2Vec (Mikolov et al., 2013) trained with gensim (Řehůřek and Sojka, 2010) on Yorùbá Wikipedia plus the training tweets (2.25M tokens, normalised like our input), frozen or fine-tuned. The fastText vocabulary contains no tone-marked under-dotted vowels (it has *won* and *wón* but never *wón*), so we look up each word exactly, then without tones, then without diacritics, raising token coverage from 80.8% to 92.2% (our Word2Vec: 96.2%). Training: Adam ($1r \cdot 10^{-3}$), batch 64, dropout 0.3, gradient clipping 1.0, early stopping on clean-dev macro-F1 (patience 3).

Transformers (E3–E5). We fine-tune mBERT, XLM-R-base, AfriBERTa-large, and AfroXLMR-base/large (Table 2) with a linear classification head using Transformers (Wolf et al., 2020) and PyTorch (Paszke et al., 2019): max length 128, AdamW with 10% linear warm-up and weight decay 0.01, learning rate $2\text{--}3 \times 10^{-5}$ for base models and 10^{-5} for AfroXLMR-large (effective batch 32), fp16, up to 10 epochs with early stopping (patience 3) on clean-dev macro-F1. For LoRA (Hu et al., 2022; Mangrulkar et al., 2022) the frozen weight matrices of the attention query/value projections are adapted as $W = W_0 + \frac{\alpha}{r}BA$ with $r = 16$, $\alpha = 32$, dropout 0.1; only A , B and the head are trained (2.6M of 560M parameters), with learning rate 10^{-4} (a rate of 3×10^{-4} collapsed two of three seeds to a single class).

Ablations (E6–E9). **E6** trains the main baseline and AfriBERTa on each diacritic form, on *mixed* (original + no-diacritics copies of every tweet) and on *mixed3* (all three forms), and evaluates every model on all three forms. **E7** adds Hausa, Igbo and Pidgin (or Hausa and Igbo only) training data

Model	Params	Yorùbá in pre-training
mBERT	178M	yes (Wikipedia)
XLM-R-base	278M	no
AfriBERTa-large	126M	yes
AfroXLMR-base	278M	yes
AfroXLMR-large	560M	yes

Table 2: Pre-trained encoders: mBERT (Devlin et al., 2019), XLM-R (Conneau et al., 2020), AfriBERTa (Ogueji et al., 2021), AfroXLMR (Alabi et al., 2022).

to AfriBERTa. **E8** uses inverse-frequency class weights. **E9** trains TF-IDF and AfriBERTa on random 10/25/50% subsets.

Evaluation protocol. Primary metric: macro-F1; we also report weighted-F1 (the official SemEval metric), per-class F1, and 95% percentile-bootstrap confidence intervals (1,000 resamples). Neural models use three seeds (mean $\pm$ s.d.). All selection uses clean dev; the test set was used once, after selection, by re-training each chosen configuration with the same seeds. Model comparisons on test use the paired bootstrap (Koehn, 2004).

Infrastructure. GPU training ran on Kaggle T4 GPUs (about 10 GPU-hours in total). Each experiment is launched from a configuration file as a notebook that clones the code repository at the exact commit and saves the executed notebook with its outputs. Engineering issues found and fixed during the project—silent multi-GPU data parallelism doubling the batch, a half-precision checkpoint loaded as fp16 master weights, and evaluation batches reordered by length grouping—are documented in the repository; every run now records whether the final model reproduces its best dev score.

5 Experiments and Results

Baselines. Word and character n-grams are complementary (Table 3): their union (E1c, 0.722, 95% CI [0.701, 0.744]) beats either alone. Training on raw tweets instead of the normalised format (E1d) costs 3.9 points, confirming that the format mismatch matters.

Recurrent models. No RNN reaches the baseline (best 0.699). Every run peaks after 2–3 epochs and then overfits, and word-level models map the many unseen test words to <unk>, whereas character n-grams still match them. Pre-trained embeddings help modestly (+1.1 to +1.8 over random). Orthography matters most: our Word2Vec, trained

ID	Model / setting	macro-F1
E0	Majority class	.198
E1a	TF-IDF word + LR	.706
E1b	TF-IDF char + linear SVM	.714
E1c	TF-IDF word+char + LR (baseline)	.722
E1d	E1c trained on raw tweets	.684
E2a	BiLSTM+attn, fastText frozen	.590 $\pm$.004
E2e	BiLSTM+attn, random init	.676 $\pm$.010
E2f	BiLSTM (no attn), fastText	.679 $\pm$.008
E2c	BiGRU+attn, fastText	.684 $\pm$.008
E2b	BiLSTM+attn, fastText	.687 $\pm$.013
E2d	BiLSTM+attn, own Word2Vec	.694 $\pm$.007
E2g	E2d + stronger regularisation	.699 $\pm$.012
E3b	XLM-R-base	.650 $\pm$.006
E3a	mBERT	.674 $\pm$.006
E5a	AfroXLMR-base	.687 $\pm$.004
E5c	AfroXLMR-large + LoRA	.708 $\pm$.004
E5b	AfroXLMR-large full FT	.714 $\pm$.029
E4	AfriBERTa-large	.731 $\pm$.006
E7a	E4 + hau, ibo, pcm	.716 $\pm$.003
E7b	E4 + hau, ibo	.715 $\pm$.009
E8	E4 + class weights	.729 $\pm$.012

Table 3: Clean-dev macro-F1 (original text; mean $\pm$ s.d. over 3 seeds for neural models).

on text normalised like the input, beats fastText and is the most robust to missing diacritics; frozen fastText fails (0.590), and frozen Word2Vec (0.657) shows that about 6.7 of those points are due to fastText’s orthography mismatch. Attention and the LSTM/GRU choice change results by less than the seed variation.

Transformers and RQ1. XLM-R-base, which saw no Yorùbá, is the weakest Transformer (0.650); AfroXLMR-base—the same architecture and tokenizer adapted to African languages—gains 3.7 points, and mBERT, whose Wikipedia data include Yorùbá, also beats XLM-R. The best model is the smallest Africa-centric one, AfriBERTa-large (0.731), ahead of AfroXLMR-large (0.714), mirroring the ranking reported with the dataset (Muhammad et al., 2023a). Full fine-tuning of AfroXLMR-large is unstable (s.d. 0.029; one seed nearly collapsed, 0.198 at epoch 2), as documented for large encoders (Mosbach et al., 2021); LoRA reaches a similar mean with 0.47% of the trainable parameters, $7\times$ lower variance and about half the time per epoch.

RQ2: diacritic robustness. Models trained on original text are brittle (Figure 2): removing tone marks at test time costs the baseline 10.3 points and AfriBERTa 6.2; removing all diacritics costs 10.2 and 7.3. Each model trained on a sin-

E6: clean-dev macro-F1 by training and evaluation diacritic form (mean over seeds)

	TF-IDF word+char + LR			AfriBERTa-large		
	original	no tones	no diacritics	original	no tones	no diacritics
trained on original	0.722	0.620	0.620	0.731	0.669	0.658
trained on no tones	0.643	0.716	0.678	0.667	0.721	0.691
trained on no diacritics	0.597	0.680	0.717	0.626	0.681	0.713
trained on mixed (orig + no diac.)	0.733	0.686	0.718	0.732	0.696	0.708
trained on mixed3 (all three)	0.732	0.720	0.717	0.723	0.716	0.713
	original	no tones evaluated on	no diacritics	original	no tones evaluated on	no diacritics

Figure 2: E6: clean-dev macro-F1 by training form (rows) and evaluation form (columns), mean over seeds.

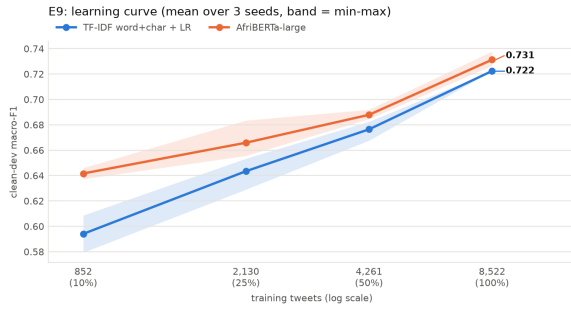


Figure 3: E9: clean-dev macro-F1 vs. training-set size (3 seeds; band = min-max).

gle form is best only on that form. Training on all three forms (*mixed3*) gives near-uniform performance—0.732/0.720/0.717 for TF-IDF and 0.723/0.716/0.713 for AfriBERTa on original/no-tones/no-diacritics—with no meaningful loss on original text. AfroXLMR models are intrinsically robust to tone removal (-0.6 to -1.4) but still lose 4–6 points without under-dots.

RQ3 and other ablations. Adding Hausa, Igbo and Pidgin data lowers Yorùbá performance to 0.716, and removing Pidgin (whose labels are skewed) does not help (0.715): diluting Yorùbá to about 30% of the training mix, rather than the label prior, causes the negative transfer. Class weighting has no effect (0.729): the imbalance is mild and negative-class errors stem from ambiguity (§6).

Learning curve. Figure 3 explains why Transformers do not pull clearly ahead: AfriBERTa’s advantage is largest with little labelled data (+4.8 points at 852 tweets) and shrinks to +0.9 at 8,522 tweets, as character n-grams catch up. Neither curve has plateaued.

Test set. Table 4 confirms the dev findings: the model ranking is unchanged and dev-to-test differ-

Model	mF1 (clean)	wF1	Δ tone	Δ diac
Majority	.197	25.3	–	–
AfroXLMR-L + LoRA	.702 $\pm$.001	73.2	–.001	–.034
BiLSTM+attn (E2g)	.710 $\pm$.005	74.7	–.075	–.085
AfriBERTa-large	.732 $\pm$.005	76.7	–.038	–.070
TF-IDF (E1c)	.737	77.2	–.081	–.084
AfriBERTa, mixed3	.741 $\pm$.007	77.5	–.002	–.012
TF-IDF, mixed3	.745	77.8	+ .001	–.006
<i>Published, same test set (weighted-F1):</i>				
AfriBERTa-large ^a	–	72.9		
AfroXLMR-large ^a	–	74.1		
SemEval-2023 best ^b	–	80.2		

Table 4: Test results: clean-test macro-F1, all-test weighted-F1 ($\times 100$, comparable to published numbers) and change in clean-test macro-F1 without tone marks / diacritics. ^aMuhammad et al. (2023a); ^bMuhammad et al. (2023b).

ences are small (-0.5 to $+1.5$ points). The two *mixed3* models are best and lose at most 1.2 points on undiacritised input, against 7–8 points without augmentation. No model significantly outperforms the TF-IDF baseline (paired bootstrap: best $p = 0.055$ for TF-IDF mixed3; $p = 0.13$ – 0.87 for the AfriBERTa mixed3 seeds). Our single models exceed the dataset’s published single-model baselines by 3–5 weighted-F1 points, plausibly in part because we train on the same format as the test set; the SemEval winners used additional pre-training and ensembles.

6 Error Analysis and Discussion

We analyse AfriBERTa (seed 42) against TF-IDF on clean test, using only reproducible evidence for quantitative claims.

Where errors occur. Confusions are spread across all class pairs (most frequent: positive $\rightarrow$ neutral, 219), and the negative class

Slice (clean test)	n	AfriBERTa	TF-IDF
All	4,129	.753 acc	.757 acc
Code-switched (vs. not)	556	.690 / .744	.696 / .743
No diacritics (vs. with)	813	.693 / .741	.681 / .741
Negation <i>kò / kì í</i>	439	.692 acc	.663 acc
Religious/greeting words	463	.911 acc	.909 acc
Negative-content words	86	.744 acc	.744 acc

Table 5: Error slices (macro-F1 unless marked acc = accuracy).

has the lowest F1. **64% of AfriBERTa’s errors (651 of 1,021) are shared with TF-IDF**, a hard core neither representation solves; each model fixes about 360 of the other’s errors, so they are partly complementary. Table 5 reports objective slices: code-switched tweets and undiacritised tweets cost 5–6 points, and negation (*kò*, *kì í*) drops accuracy to 0.69 (TF-IDF 0.66) against 0.75 overall.

Keyword priors. 84% of tweets containing religious or greeting words (*Ọlórún*, *Olúwa*, *àdúrà*, *ẹ kú*) are positive, so models learn “God $\Rightarrow$ positive”: half of their errors on such tweets (20/41) are false positives, e.g. a Bible verse (*Ní àtètèkòşẹ Ọlórún dá ọrun àti ayé*, gold neutral). Symmetrically, 77% of errors on tweets with words such as *ikú* ‘death’ or *àrùn* ‘disease’ follow the negative word, e.g. positive news about a rescue from Boko Haram.

Label noise. Two reproducible signals indicate that part of the remaining error lies in the labels. First, 15 of the 386 test tweets that duplicate a training tweet carry a *different* gold label, and AfriBERTa predicts the training copy’s label in all 15 (one is a condolence message, labelled negative in train but positive in test). Second, confident learning (Northcutt et al., 2021) with out-of-fold TF-IDF probabilities flags 10.8% of training tweets (14.0% of negatives) and 9.4% of test tweets as likely label issues, consistent with the moderate inter-annotator agreement ($\kappa = 0.65$). Flags are candidates, not proof, but they suggest roughly 10% of the error is not reachable by any model trained on these labels.

Idioms, proverbs and calibration. Reading the sampled errors (model-assisted glosses, not verified by a native speaker, so used only illustratively) shows sentiment carried by idioms and proverbs: *àkàrà tú sẹpo* (‘the bean cake has dissolved in the oil’, i.e. things fell apart) is negative but predicted

neutral; *bójú bá farabalẹ á rímú* (‘if the eye is patient it will see the nose’) is positive but predicted negative. Educational posts that quote a proverb are labelled by the proverb’s content, the model often by the post’s neutral style. AfriBERTa is also over-confident: 94% of its test predictions have probability above 0.9, but only 77% of those are correct (Guo et al., 2017).

Discussion. With 8.5k noisy, short tweets, a well-tuned character-aware n-gram model is a strong competitor to pre-trained Transformers; pre-training pays off most when labels are scarce, and African-language pre-training data matters more than model size, consistent with Ogueji et al. (2021) and Muhammad et al. (2023a). The practically most important result is cheap: rendering training data in all written forms makes both model families robust to how Yorùbá is actually typed.

7 Deployment

The deployed system (text input $\rightarrow$ training-identical normalisation $\rightarrow$ two classifiers $\rightarrow$ class probabilities) is a Streamlit application (Streamlit, 2019) on Streamlit Community Cloud. A user types or pastes a tweet, optionally simulates typing without tone marks or diacritics, and sees both mixed3 models’ predicted labels and class probabilities side by side, the normalised text the models receive, and a warning when the models disagree. Example buttons include known hard cases. Input goes through exactly the training preprocessing code. The fine-tuned AfriBERTa (seed 42) and the TF-IDF model are published on the Hugging Face Hub with a model card. Because free Hugging Face hosting of Python applications now requires a paid plan and the free Streamlit tier offers about 1 GB of memory, AfriBERTa is served as a per-channel int8 ONNX model with ONNX Runtime (ONNX Runtime developers, 2018) (127 MB, no PyTorch): on clean dev it agrees with the original model on 96.3–96.6% of tweets and changes macro-F1 by at most 0.3 points; tokenisation is identical. The application uses about 550 MB of memory and answers in about 0.05 s per tweet.

8 Limitations and Future Work

Data: one domain (Twitter), three coarse classes, about 10% suspected label noise and $\kappa = 0.65$; no dialectal or Benin Yorùbá coverage. **Evaluation:** the test set is small for distinguishing models a few points apart; meaning-based error cate-

gories were not verified by a native speaker. **Models:** no Transformer significantly beats TF-IDF; probabilities are over-confident; negation and idioms remain unsolved. **System:** the free host sleeps after inactivity. **Future work:** native-speaker re-annotation of flagged tweets; language- and task-adaptive pre-training as used by the best SemEval systems (Wang et al., 2023); ensembling TF-IDF and AfriBERTa, whose errors are partly complementary; negation-aware augmentation; temperature-scaling calibration; and comparison with diacritic restoration (Orife, 2018) as an alternative to augmentation.

9 Conclusion

We built and deployed a Yorùbá tweet sentiment classifier and found that the data, not the model, set most of the challenges: a hidden train/test format mismatch, duplicate leakage, label noise and diacritic variation. Africa-centric pre-training helps (RQ1), but on 8.5k tweets a well-tuned TF-IDF model is statistically indistinguishable from the best Transformer. Models trained on diacritised text are brittle to undiacritised input, and training on all written forms fixes this at no cost (RQ2). Data from related Nigerian languages did not help (RQ3). The main lessons are to audit the data before modelling, to evaluate with multiple seeds and significance tests, and to evaluate under the conditions in which a system will actually be used.

Use of Existing Resources and Own Contributions

Existing resources used: the AfriSenti-SemEval 2023 data (Muhammad et al., 2023a); the pre-trained models mBERT, XLM-R, AfriBERTa and AfroXLMR; fastText Yorùbá vectors (Grave et al., 2018); Yorùbá Wikipedia text; and open-source libraries (PyTorch, Transformers, PEFT, scikit-learn, gensim, ONNX Runtime, Streamlit). **Implemented in this project:** the data pipeline (pinned loading, reconstruction and validation of the test normalisation, duplicate detection, diacritic transformations); all training, evaluation and analysis code (baselines, BiLSTM/BiGRU with attention, Word2Vec training and tiered embedding lookup, Transformer and LoRA fine-tuning, diacritic augmentation, bootstrap statistics, learning curves, error slices, confident-learning analysis); the reproducible Kaggle experiment pipeline; all 30+ experiments; and the deployment

(model export and quantisation, validation, web application).

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